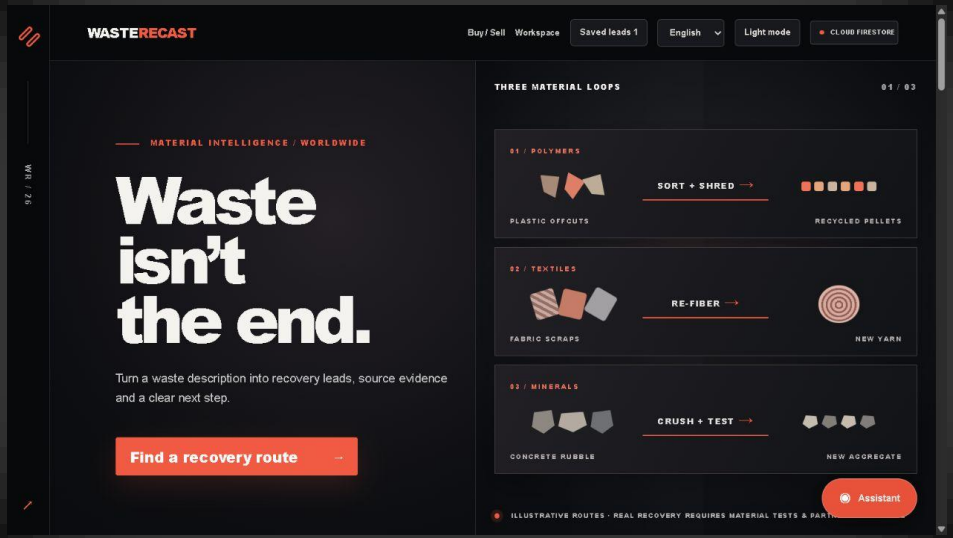
Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details
Team name: Clorex | Product: WasteRecast
Team leader: Zuhaib Ahmed
Members: Anshuman Pandey · Aliyan Ahmed
Problem statement: Sustainability & Social Impact
Evidence-based material reuse and global matching



WasteRecast

Find a counterpart. Check acceptance conditions. Work through what needs to change.

Built by Clorex

Brief about the idea

A category match does not establish material acceptance.

THE BUSINESS PROBLEM

Generators and buyers describe the same material differently. A promising listing can still fail on moisture, contaminants, grade or usable cost.

OUR WORKING SOLUTION

A global exchange retrieves possible counterparts, checks sourced specifications and explains a supported change to investigate before a trade.

Theme: Sustainability & Social Impact — resource efficiency and informed decisions. No completed recovery or emissions savings claimed.

Opportunities

How is this different?

Search retrieves possibilities by meaning. Typed checks then distinguish a stated fit, an unknown fact, a failed condition and an incompatible material.

How will it solve the problem?

Translate evidence into comparable units. Investigate a documented process for a partial fit. Recalculate economics from quotes and measured yield.

Our proposed USP

A smallest-supported-change investigation explains what a partial match needs. It preserves the failed condition until a trial and retest support acceptance.

Existing exchanges also match materials. Our proposed distinction is the combined evidence, constraint and conditional-processing workflow. Pilot validation is pending.

List of features offered by the solution

01 / GLOBAL REGISTERED EXCHANGE

Verified Google/email accounts; buyer requests and seller offers; country filters; owner controls; 30-day expiry.

02 / HYBRID RETRIEVAL

Alias/IDF overlap + actual 768-dimensional Gemini vectors + curated process edges, fused by reciprocal rank.

03 / ACCEPTANCE PASSPORT

12 typed facts with quotations, references and review state. Compatible-unit checks return pass, fail or unknown.

04 / DECISIONS AND FOLLOW-THROUGH

Conditioning investigations, sourced landed costs, private review/outcomes; public recovery directory and Recast assistant.

Vector search requires verified sign-in. Missing vectors trigger a labelled alias/process fallback. Public directory listings are not registered users.

Process flow diagram or Use-case diagram

1 Describe

Offer or requirement
Optional evidence

→

2 Retrieve

Aliases + vectors
Process edges / RRF

→

3 Check

Typed pass/fail/unknown
Gemini explanation

→

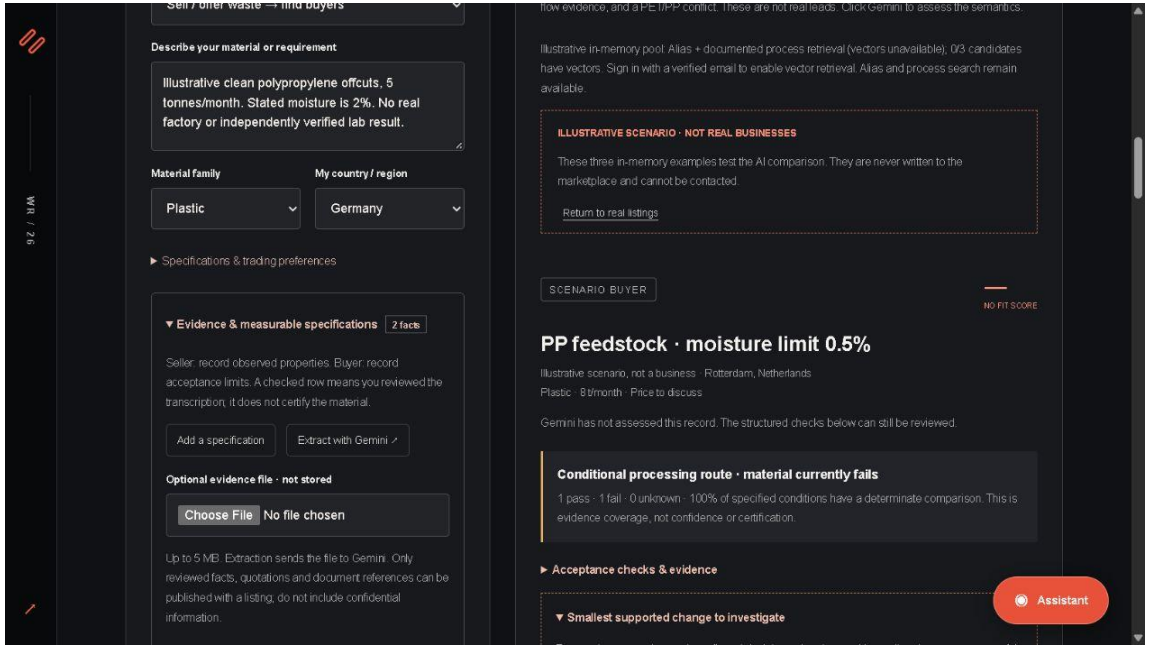
4 Investigate

Supported operations
Quotes + usable yield

5 Confirm with the counterpart
Sample trial and retest → current authorization → written acceptance → commercial terms

AI interprets and explains; deterministic checks and calculations remain inspectable. A human confirms acceptance before any movement.

Wireframes/Mock diagrams of the proposed solution (optional)

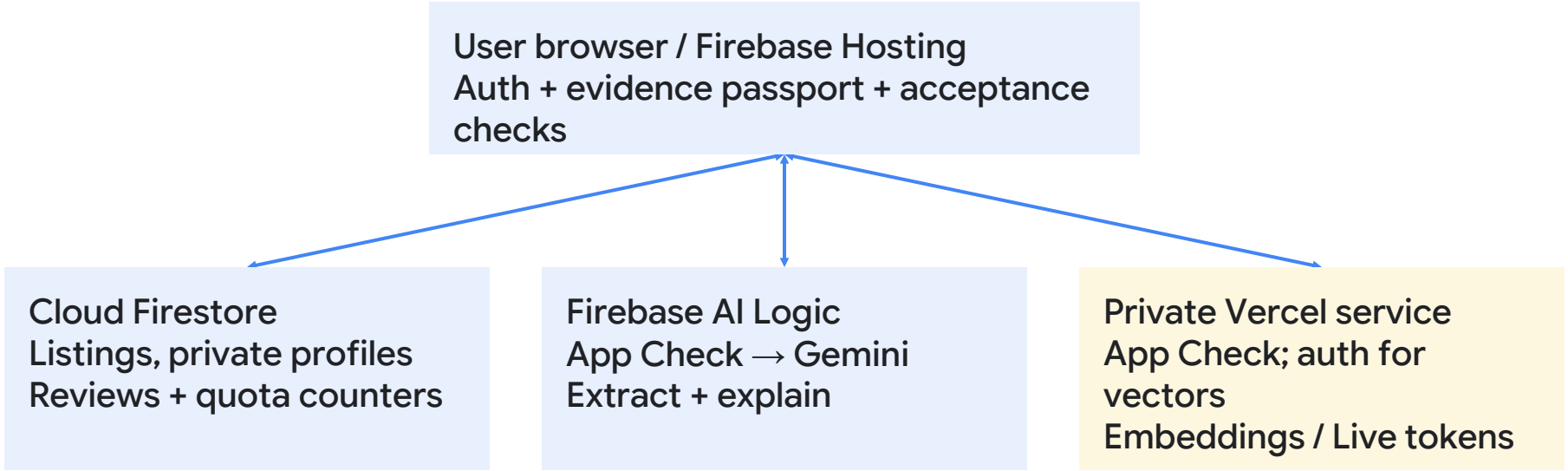


ACTUAL INTERFACE

1. Compare meaning
2. Inspect each condition
3. Resolve evidence gaps

Actual deployed interface. Illustrative material and counterpart specifications. Facts retain review and source status.

Architecture diagram of the proposed solution



Actual vectors + aliases + process edges → rank fusion → typed checks → conditional investigation.
Financial feasibility uses entered quotes and usable yield, separately from semantic fit.

Firebase is the deployed hosting path. Cloud Run is not deployed. API keys stay server-side; model and database free quotas apply.

Technologies to be used in the solution

Firebase Hosting + Auth	Public HTTPS app; Google/email sign-in; verified owners manage exchange listings.
Cloud Firestore	Global offers/requests, consented business contacts, private reviews and durable embedding quotas; sourced directory.
Gemini + Firebase AI Logic	Evidence extraction and constrained semantic explanations. App Check protects public requests.
Gemini embeddings + Live	768-dimensional semantic vectors; private Vercel endpoint; one-use Live voice tokens.
Application logic	Alias/IDF + cosine + RRF retrieval; typed unit checks; curated process solver; referenced landed-cost calculation.

No Cloud Run, Cloud SQL, BigQuery, paid vector database or Vertex AI deployment. Firebase is the chosen permitted hosting path.

Estimated implementation cost (optional)

Current prototype: configured for free tiers.

CURRENT SETUP

Firebase Spark + free Gemini quota + existing Vercel project. No Google Cloud billing account linked.

SCALING APPROACH

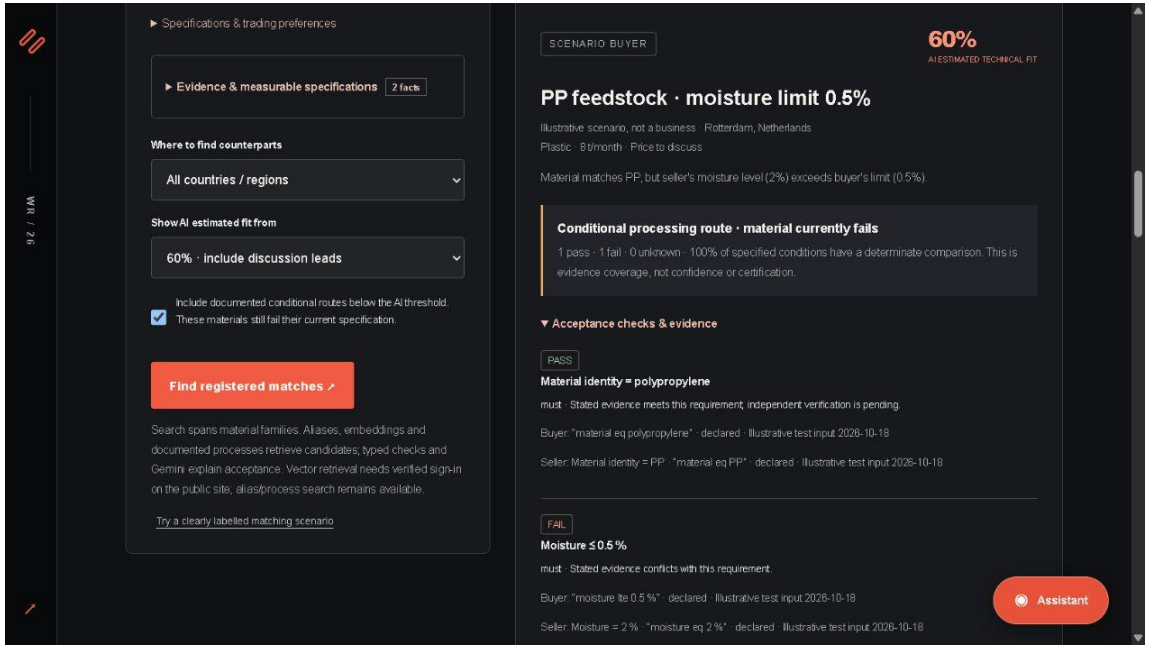
Static delivery; bounded 200-candidate initial pool; vector cache; durable 240-text/account/24-hour embedding budget; transparent fallback.

Before a paid rollout

Measure traffic and model usage. Add server retrieval indexes, durable vector caching, moderation and monitoring before broad rollout.

No cost or unlimited-availability guarantee. A paid production estimate requires measured traffic and current provider pricing.

Snapshots of the prototype



PARTIAL FIT → NEXT STEP

PP identity: pass

Moisture: fail

Drying: investigate, then retest

Actual public-app capture. Illustrative PP facts and buyer requirements. A conditioning suggestion does not establish successful acceptance.

Prototype Performance report/Benchmarking

57 AUTOMATED TESTS PASS

Regression tests cover retrieval, units, conflicting evidence, unknowns, economics, security and existing voice controls.

24 AUTHORED ACCEPTANCE CASES

24/24 expected decisions. Tests include missing facts, contradictions, hard conflicts and process conditions. Expert labels pending.

LIVE SERVICE CHECKS

29 cloud/access checks and 13 Firebase SDK checks pass. Actual 768-dimensional vectors and public Gemini explanations tested.

LIMITED LIVE EVALUATION

One scenario, 3 model runs: conditional in all three; scores 55/70/65. Two 31-record retrieval fixtures: target ranked first.

18 October 2026. Authored fixtures and service tests are not market accuracy. No independent expert study, load benchmark or verified trade result.

Additional Details/Future Development (if any)

NEXT / VERIFIED PILOT

Onboard consenting buyers and sellers. Validate evidence, accepted specifications, sample outcomes and written commercial terms.

PILOT / PROVE THE VALUE

Measure time to qualified enquiry, acceptance rate and verified recovered mass with real generator–recycler pairs.

SCALE / OPERATE RELIABLY

Server-side retrieval indexes, shared vector caching, abuse controls, monitoring and dated regional source refreshes.

EXPAND / QUOTED ECONOMICS

Road-route distances, buyer-specific logistics quotes and auditable price comparisons. These are not live today.

Initial search coverage is bounded, with manual pagination. No claim of exhaustive global retrieval, perfect matching or measured environmental benefit.

Provide links to your:

FINAL PRODUCT LINK

https://wasterecast.web.app/

GITHUB PUBLIC REPOSITORY

https://github.com/anshuman542018/wasterecast

UPDATED DEMO VIDEO / 3 MINUTES

https://drive.google.com/file/d/1e1EyM-4qKs6WXAbR9nq35Z9UsCGghN5u/view

All links must be accessible without requesting access. Submit the PDF exported from this prescribed template.

Evidence, boundaries and evaluation readiness

Every proposed trade needs evidence and acceptance.

Two data sources	Registered listings are self-declared. The separate 912-record India directory and worldwide map leads need verification.
AI and fixed checks	Fixed checks preserve failures and unknowns. Gemini fit scores are uncalibrated estimates, never acceptance certificates.
Conditioning and cost	Processing needs a trial, retest and written acceptance. Costs require references; no yield or FX is invented.
Evaluation access	Public app and source. Free quotas apply. Expert validation and verified commercial outcomes remain pending.

Document uploads use Google free-tier processing terms. Business contacts need publication consent. Private profiles/reviews have owner-only access.

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Thank you

